RUPAM SEN

RTL Design Engineer - Vervesemi Microelectronics
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PROFESSIONAL SUMMARY

ASIC Design Engineer with 2.5 years of experience in RTL design and Digital IP development, contributing to 2 successful tape-outs as the primary RTL owner of Vervesemi's first motor control MCU. Proficient in micro-architecture definition, emulation-compatible RTL design, and implementation of motor-control and communication protocol IPs. Skilled in cross-team collaboration and design bring-up across full chip development cycles.

WORK EXPERIENCE

- **VERVESEMI MICROELECTRONICS -** *Digital Design Engineer II* — *Oct 2023 - Present*

  MCPWM - MOTOR CONTROL PULSE WIDTH MODULATION
  - Designed and implemented the RTL in Verilog for the **MCPWM module**, generating up to 12 independent PWM outputs with core features including dead-time generation, fault detection, and output override control.
  - Performed **IP-level functional verification** using SystemVerilog testbenches over an **AXI-Lite interface**; developed **C-based environments** for SoC-level verification using **Cadence Xcelium**.
  - Authored synthesis constraints in **SDC format**, synthesized the RTL using **Cadence Genus**, and validated design functionality through post-synthesis simulation and static timing analysis.

  CORDIC COPROCESSOR
  - Developed **CORDIC Coprocessor** to compute trigonometric and hyperbolic functions (sin, cos, tanh, arctan, square root, and modulus) using the **CORDIC algorithm and Digit-by-digit method.**
  - Achieved **up to 15-bit precision** on 16-bit outputs, and optimized pipeline architecture for **improved performance and area efficiency.**
  - Developed an **APB interface-based** environment for IP-level functional verification. Performed SoC-level verification and Gate-level simulation.

  HALL SENSOR INTERFACE
  - Implemented **datapath and FSM-based control logic** for **Hall sensor interface** block to capture the BLDC motor's rotor position and speed.
  - Developed interpolation logic to generate a **high-resolution 16-bit rotor angle** from low-resolution Hall sensor inputs.
  - Supported **FPGA-based validation** and performed real-time signal debugging using **Integrated Logic Analyzer** in **Vivado.**

- **CADENCE DESIGN SYSTEMS -** *Product Validation Intern* — *Aug 2022 - July 2023*
  - Experienced on the **SV-AMS**, **Wreal**, and **RNM technique**s. Designed AMS testcases to validate **Xcelium**.
  - Conducted daily **regression analysis.** Identified the root cause of the failed regression testcases and debugging. Worked with the version control system **- Perforce.**

PROJECTS

- **RISC-V RV32I Pipelined Processor Core |** Verilog/SystemVerilog, Cadence Xcelium
  - Designed and implemented a **5-stage pipelined RV32I processor** core in Verilog. Developed a SystemVerilog testbench with self-checking scoreboard, and validated the design by running assembly programs [i.e. **Fibonacci** and **Bubble Sort**].

TECHNICAL SKILLS

- **Languages:** Verilog, SystemVerilog, UVM, Verilog-AMS, C/C++, Python, Perl, TCL, Makefile
- **Verification:** Lint, CDC, Assertion, Coverage, UPF, Synthesis, Gate-level Simulation
- **Interfaces:** AXI, APB, UART, I2C, SPI

EDUCATION

- **Master of Technology** [Communication Systems] — CGPA: **9.1**
  Motilal Nehru National Institute of Technology (**MNNIT**), Allahabad — **2023**
- **Bachelor of Technology** [ECE] — CGPA: 8.76
  University of Calcutta – Institute of Radio Physics and Electronics (**IRPE**), Kolkata — 2021